IoT Lab 4 Report

Abstract

This project presents a smart Automatic Number Plate Recognition (ANPR) system using a Raspberry Pi connected with a camera module. The system captures live video, detects vehicle number plates using computer vision techniques, and recognizes plate characters using OpenCV and Tesseract OCR. The recognized number is displayed in real time and stored in a text file for future use. This setup demonstrates a low-cost embedded surveillance and automation solution for smart parking, security gates, and traffic monitoring. Inspired by the structure of the uploaded IoT hardware report.

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1. Introduction

Automatic Number Plate Recognition is an important application in modern intelligent transportation systems. It is used in toll booths, parking systems, law enforcement, and access control. Traditional high-end ANPR systems are expensive and require dedicated hardware.

This project implements ANPR using a Raspberry Pi and a camera module. The Raspberry Pi processes live video streams, detects plate regions, extracts characters, and stores the recognized data. By combining embedded hardware with image processing and OCR, the system provides an affordable and scalable solution.

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2. System Architecture

2.1 Hardware Components

The setup consists of:

- Raspberry Pi 4
- Raspberry Pi Camera Module with CSI
- MicroSD Card with OS installed
- Power Supply
- HDMI Monitor

2.2 Software Components

- Raspberry Pi OS
- Python

- OpenCV
- Tesseract OCR
- Time and File Handling Libraries

2.3 Working Block Diagram

Camera Input
↓
Live Video Stream
↓
OpenCV Processing
↓
Number Plate Detection
↓
Tesseract OCR
↓
Recognized Plate Text
↓
Display + Save to File

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3. Hardware Setup Procedure

Step 1: Camera Connection

The camera is connected to the Raspberry Pi using the CSI port (for Pi Camera).

Step 2: Enable Camera

Camera interface is enabled in Raspberry Pi settings.

Step 3: Power On System

The Raspberry Pi boots into the operating system. **Step**

4: Install Required Packages

sudo apt update sudo apt install python3-opencv tesseract-ocr
numpy pytesseract

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4. Methodology

4.1 Live Video Capture

The camera continuously captures frames. The frames are streamed using Raspberry Pi camera commands and read in Python using OpenCV.

4.2 Number Plate Detection

Using OpenCV image processing:

- Convert frame to grayscale
- Apply edge detection
- Find contours
- Select rectangular regions matching plate shape Bounding boxes are drawn around detected plates.

4.3 Character Recognition

The processed plate image is passed to Tesseract OCR.

Tesseract uses LSTM-based neural networks internally to convert image pixels into text.

Example Output:

BR01AB1234

4.4 Data Logging

Recognized number plates are stored in: **plates.txt**

This can later be connected to databases, gate systems, or alerts.

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5. Results and Observations

The system successfully performs:

- Real-time live video capture
- Number plate detection using OpenCV
- Character recognition using Tesseract OCR
- Plate display on screen
- Automatic saving to file

Sample Output

PLATE: BR01AB1234
PLATE: JH05XY9876

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6. Challenges Faced

- Low light reduces camera clarity

- Motion blur at high speed vehicles
- Small plates difficult to detect
- Two-line plates require extra OCR handling
- Incorrect OCR for unclear fonts
- CPU limitations on Raspberry Pi for heavy AI models

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7. Improvements / Future Work

- Use quantized deep learning OCR models
- Use YOLO Nano for better plate detection
- Connect to cloud database
- Add gate automation system
- Real-time blacklist checking
- Mobile notifications on detection

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8. Code Link:

https://drive.google.com/file/d/1QKtyj50XuPmQdpRnzf28uipOKzHWrofo/view?usp=sharing

Command to run:

- rpicam-vid -t 0 --width 640 --height 480 --framerate 25 --inline --listen --flush --nopreview -o tcp://0.0.0.0:8888
- python hardware_4.py

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9. Conclusion

This project demonstrates an efficient and low-cost ANPR system using a Raspberry Pi and camera. By integrating OpenCV for detection and Tesseract OCR for recognition, the system provides a practical solution for smart surveillance and access control. It highlights the power of embedded systems in solving real-world automation problems.